Home Lab 3: The Starch and Carbonate Tests

Do this with a parent. Goggles on. Never taste anything. Clean up after.

Goal

See exactly what the iodine test and the acid-fizz test do — including the tricky results (powdered sugar, baking powder, vitamin C).

You need

From the kit: **corn starch, granulated sugar, powdered sugar, calcium carbonate (chalk), baking soda**; iodine. From home: baking **powder**, a crushed vitamin C tablet (or a pinch of the vitamin C from the kit if included) dissolved in a little water, white vinegar, water, toothpicks, cups, paper towels.

Part 1: Iodine

Put a dry pinch of each on a paper towel. Add **one drop of iodine** to each. **Predict first**, then test.

Powder	My prediction	Actual (blue-black / orange / colorless)
Corn starch		
Granulated sugar		
Powdered sugar		
Baking powder		
Baking soda		
Vitamin C solution (drop iodine into it)		

Part 2: Acid fizz

Put a fresh pinch of each on a paper towel and add a few drops of vinegar.

Powder	Fizz?	How strong?
Baking soda		
Calcium carbonate (chalk)		
Corn starch		
Baking powder (add water first, then vinegar)		

Follow-up

1. Powdered sugar gave a faint blue with iodine even though sugar has no starch. Why?
2. Baking powder turned blue-black but baking soda didn't. What does baking powder contain that baking soda doesn't?
3. What happened when you added iodine to the vitamin C solution? Is that the same kind of thing as the starch test?
4. Chalk and baking soda both fizzed. Which fizzed faster, and why?